Machine Learning-Driven Compact Modeling of Advanced Devices

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Abstract— This research develops a machine-learning framework for compact modeling of advanced semiconductor devices using TCAD-generated data. With 10,000 datasets of key parameters like VG, and ID, data extraction and K-means clustering streamline training. Among evaluated methods, a Multi-Layer Perceptron (MLP) achieved high accuracy with an MSE below 0.02. The integration of TCAD simulations and machine learning highlights an efficient approach for scalable, accurate semiconductor device modeling.

Introduction

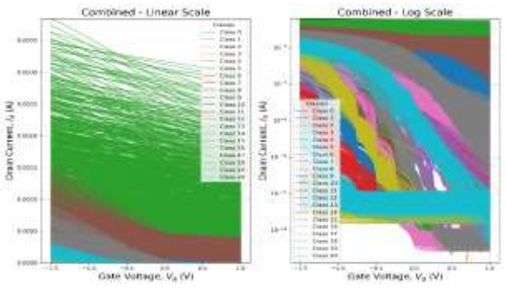
Traditional compact modeling approaches, such as physics-based models (e.g., BSIM) and empirical methods, are accurate but computationally intensive, making them unsuitable for large-scale or iterative circuit simulations [1] [2]. As semiconductor devices grow more complex, these methods become time-consuming and require significant expertise to adapt to emerging materials like GaN.

Machine learning offers a transformative solution, accelerating the development of compact models by automating parameter fitting and reducing simulation time. Neural networks, in particular, provide scalable and accurate predictions, with Multi-Layer Perceptron (MLP) models achieving up to 40% faster simulation speeds compared to traditional methods [2]. This project integrates TCAD simulations with machine learning, streamlining the generation of GaN device compact models for efficient design and optimization.

Methodology

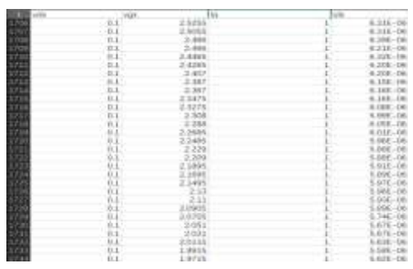
TCAD simulation & Clustering

A dataset of 10,000 samples was generated through TCAD simulations of GaN-based semiconductor devices. Critical parameters like VG, VD, and ID were extracted. K-means clustering grouped the data into clusters with similar I–V characteristics, ensuring efficient and representative machine learning training.



Preprocessing

Key features such as VG, VD, and lg parameters were selected for their importance in predicting I–V characteristics. And to enhance model performance, a logarithmic transformation was applied to ids.



ML Model(MLP) Train

A Multi-Layer Perceptron (MLP) neural network was designed with input, hidden, and output layers tailored to model GaN device behavior. The model was trained using the Adam optimizer, with hyperparameters like learning rate and batch size fine-tuned for convergence.



Evaluation & Visualization

The trained MLP was evaluated using test data to validate its performance. Key metrics like mean squared error (MSE) were calculated. The I–VI–V relationships were visualized, comparing predicted curves to simulation data to confirm the model's accuracy and to identify areas for future optimization.



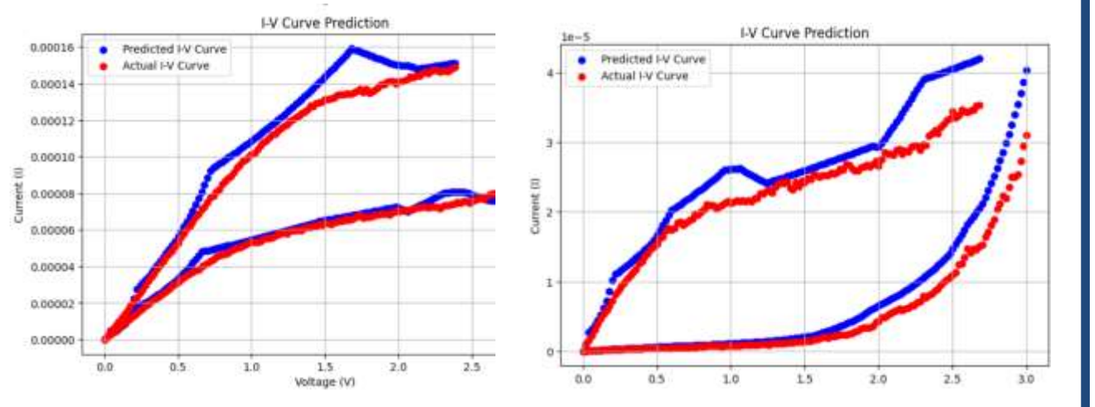
Results and Discussion

Using TCAD simulations, a dataset of 10,000 samples was generated, capturing key parameters like gate voltage (VG), drain voltage (VD), and drain current (ID). K-means clustering was applied to group datasets with similar I–V characteristics, reducing redundancy and improving the efficiency of model training.

A Multi-Layer Perceptron (MLP) neural network was trained on the clustered data and achieved a mean squared error (MSE) below 0.02, indicating high predictive accuracy. The loss curves demonstrated effective convergence, highlighting the model's ability to generalize across diverse device configurations. Compared to traditional methods, the MLP model reduced simulation time by up to 40% without sacrificing accuracy.

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train epoch[200/200] loss:0.008: 100%|██████████| 145/145 [00:00<00:00, 584.80it/s]
100%|██████████| 17/17 [00:00<00:00, 8581.44it/s]
[epoch 200] train_loss: 0.014  val_mse: 0.000721
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Visualizations of the predicted I–V curves showed close alignment with the actual TCAD data, with deviations of less than 3% in critical regions such as subthreshold and saturation. These results validate the framework's robustness and scalability for advanced compact modeling.



Conclusion and Future Work

This project has successfully established a machine-learning framework for compact modeling of advanced semiconductor devices, reducing computational time by up to 40%, validating its efficiency and scalability [3]. The clustering and modeling processes demonstrated the capability of integrating TCAD simulations with neural networks, achieving a mean squared error (MSE) below 0.02 and significant time savings compared to traditional methods.

Future work will focus on incorporating additional device parameters, such as capacitance-voltage (C–V) characteristics and transconductance, to expand the model's applicability. Exploring alternative machine learning algorithms, including Support Vector Machines and Decision Trees, will provide comparative insights and enhance the robustness of the framework. Additionally, the future work will also aim to generalize the compact model for emerging device architectures, such as GAAFETs and NCFETs.

Selected References

[1] Y. Zhang, G. He, K. -T. Tang, Y. Li and G. Wang, "GEM: A Generalized Memristor Device Modeling Framework Based on Neural Network for Transient Circuit Simulation,"

[2] C. -T. Tung, M. -Y. Kao and C. Hu, "Neural Network-Based and Modeling With High Accuracy and Potential Model Speed," in *IEEE Transactions on Electron Devices*, vol. 69, no. 11, pp. 6476-6479, Nov. 2022, doi: 10.1109/TED.2022.3208514.

[3] J. Wang, Y. -H. Kim, J. Ryu, C. Jeong, W. Choi and D. Kim, "Artificial Neural Network-Based Compact Modeling Methodology for Advanced Transistors," in *IEEE Transactions on Electron Devices*